

Financial Mathematics 32000

Lecture 5

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UNIT 3: Monte Carlo

Generating random variables

Monte Carlo estimate

Let Y be a discounted payoff. Example: $Y = e^{-\int_0^T r_t dt} (S_T - K)^+$.

Want to calculate the time-0 price $C = \mathbb{E}Y$

Generate $Y_1, Y_2, \dots$ independently and identically distributed as Y .

(How?)

Then the random variable

$$\hat{C}_M := \frac{Y_1 + Y_2 + \dots + Y_M}{M}$$

is the Monte Carlo estimate of C . Note that

$$\mathbb{E}\hat{C}_M = C$$

By the strong law of large numbers, with probability 1 we have

$$\hat{C}_M \rightarrow C \text{ as } M \rightarrow \infty.$$

How fast does convergence occur?

Let $\sigma^2 := \text{Var}(Y)$. (Here σ does not denote volatility.) Then

$$\text{Var}(\hat{C}_M) = \frac{1}{M^2} \text{Var}(Y_1 + Y_2 + \cdots + Y_M) = \frac{1}{M^2} (M\sigma^2) = \frac{\sigma^2}{M}$$

By the Central Limit Theorem, we have convergence in distribution

$$\frac{\hat{C}_M - \mathbb{E}\hat{C}_M}{\sqrt{\text{Var } \hat{C}_M}} \xrightarrow{d} N(0, 1) \quad \text{hence} \quad \frac{\hat{C}_M - C}{\sigma/\sqrt{M}} \xrightarrow{d} N(0, 1)$$

as $M \rightarrow \infty$. Conclusion still holds using sample stdev in place of σ :

$$\frac{\hat{C}_M - C}{\hat{\sigma}_M/\sqrt{M}} \xrightarrow{d} N(0, 1)$$

because $\hat{\sigma}_M^2 \rightarrow \sigma^2$ where

$$\hat{\sigma}_M^2 := \frac{1}{M-1} \sum_{m=1}^M (Y_m - \hat{C}_M)^2.$$

is the “sample variance” and $\hat{\sigma}_M$ is the “sample standard deviation.”

Confidence intervals

95%

0.05

So an asymptotic (for large M) confidence interval of $100(1 - p)\%$ is

$$\left(\hat{C}_M - \overset{1.96}{\mathcal{N}^{-1}\left(1 - \frac{p}{2}\right)} \frac{\hat{\sigma}_M}{\sqrt{M}}, \hat{C}_M + \mathcal{N}^{-1}\left(1 - \frac{p}{2}\right) \frac{\hat{\sigma}_M}{\sqrt{M}} \right)$$

where $\mathcal{N}$ is the standard Normal cdf.

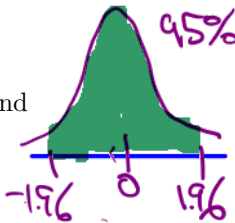
- ▶ $\mathcal{N}^{-1}\left(1 - \frac{p}{2}\right)$ tells us: a radius of *how many standard deviations* of a normal distribution contains $100(1 - p)\%$ of the probability?
- ▶ The $\hat{\sigma}_M/\sqrt{M}$ is called the *standard error*.

It gives us the estimated standard deviation of $\hat{C}_M$.

- ▶ Example: Let $p = 0.05$. Then $\mathcal{N}^{-1}\left(1 - \frac{p}{2}\right) \approx 1.96$ and

$$\left(\hat{C}_M - 1.96 \frac{\hat{\sigma}_M}{\sqrt{M}}, \hat{C}_M + 1.96 \frac{\hat{\sigma}_M}{\sqrt{M}} \right)$$

is an asymptotic 95% confidence interval for C .



Confidence intervals

- ▶ If $\hat{\sigma} = 20$ and we run $M = 10000$ simulations, then a 95% confidence interval has radius

$$1.96 \times \frac{20}{\sqrt{10000}} = 0.40$$

To reduce this to 0.04, we need to take $M = 1$ million.

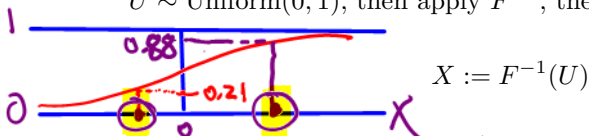
- ▶ So we will want to use *variance reduction* techniques, which reformulate the problem to keep σ small, or which carefully introduce dependence in the simulations to keep $\text{Var } \hat{C}_M$ small.

UNIT 3: Monte Carlo

Generating random variables

The inverse CDF method

- Assume the existence of a pseudo-random number generator whose output can be treated as if it is IID uniform on $(0, 1)$.
Python: `numpy.random.Generator` has method `random()`
- To generate a random variable X having a CDF F , generate $U \sim \text{Uniform}(0, 1)$, then apply F^{-1} , the inverse CDF, to produce



As desired, $\mathbb{P}(X \leq x) = \mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$.

(If F not invertible, letting $F^{-1}(u) := \min\{x : F(x) \geq u\}$ works.)

Intuition: randomly choose the *percentile* between 0% and 100%, uniformly. The inverse CDF finds the corresponding value of X .

Generating normal random variables

- ▶ Python: `numpy.random.Generator` has method `normal()`

But what if you need to build your own?

- ▶ Could do $\mathcal{N}^{-1}(U)$, if an implementation of the inverse of the normal CDF $\mathcal{N}$ is available. Excel: `NORMSINV(RAND())`

- ▶ Box-Muller method: If (X, Y) are independent $\text{Normal}(0, 1)$, then $R := X^2 + Y^2$ has CDF $\mathbb{P}(R \leq r) = 1 - e^{-r/2}$. Given R , the point (X, Y) is uniformly distributed on the circle of radius $\sqrt{R}$. So generate *pairs* of independent normals by drawing U_1 and U_2 IID from a $\text{Uniform}(0, 1)$ distribution, and taking

$$R := -2 \log(U_1)$$

$$(X, Y) := (\sqrt{R} \cos(2\pi U_2), \sqrt{R} \sin(2\pi U_2))$$

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Simulating random variables in Python

```
[1] import numpy
    rng = numpy.random.default_rng(seed=0)
```

```
[2] rng.random(size=5)

array([0.63696169, 0.26978671, 0.04097352, 0.01652764, 0.81327024])
```

```
[3] rng.random(size=5)

array([0.91275558, 0.60663578, 0.72949656, 0.54362499, 0.93507242])
```

```
[4] rng.normal(size=5)

array([-0.62327446,  0.04132598, -2.32503077, -0.21879166, -1.24591095])
```

```
[5] rng.normal(size=(2,5))

array([[ -0.73226735, -0.54425898, -0.31630016,  0.41163054,  1.04251337],
       [-0.12853466,  1.36646347, -0.66519467,  0.35151007,  0.90347018]])
```

In some cases, no need to simulate entire path

If Y is a known function of random variables with distributions that you can directly simulate, then it is easy to generate $Y_1, Y_2, \dots$

Example: Consider a call paying $(S_T - K)^+$ where S is GBM.

$$dS_t = rS_t dt + \sigma S_t dW_t$$

Then

$$Y = e^{-rT}(S_T - K)^+ = e^{-rT}(S_0 e^{(r-\sigma^2/2)T + \sigma W_T} - K)^+$$

where $W_T \sim N(0, T)$. So let

$$Y_m := e^{-rT}(S_T - K)^+ = e^{-rT}(S_0 e^{(r-\sigma^2/2)T + \sigma\sqrt{T}Z^{(m)}} - K)^+$$

where the $Z^{(m)}$ are IID standard normal: $N(0, 1)$.

But sometimes need to simulate entire path

- ▶ But what about more complicated dynamics, such as

$$dS_t = rS_t dt + \sigma(S_t, t)S_t dW_t$$

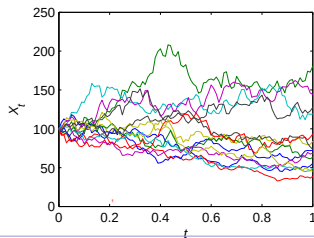
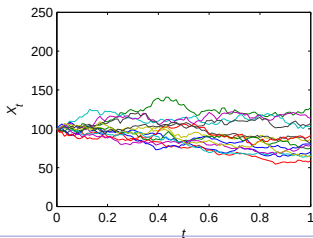
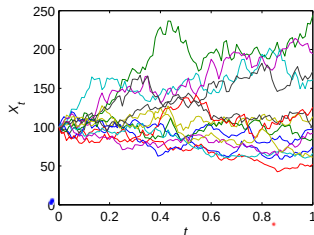
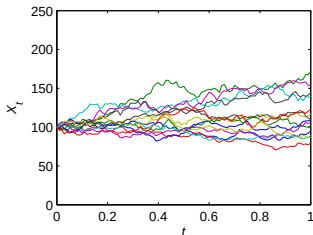
- ▶ Or what if σ or r follows a process driven by a second BM.
- ▶ What about more complicated contracts, such as an Asian option or a barrier option.

We may need to simulate the whole path.

Geometric Brownian motion: $dX_t = \mu X_t dt + \sigma X_t dW_t$

$L = \log X$ $dL_t = (\mu - \frac{1}{2}\sigma^2) dt + \sigma dW_t$

Let $X_0 = 100$. Trajectories for $\mu = -0.15, +0.15$ and $\sigma = 0.20, 0.40$:



e^L

Euler method to simulate path of a state variable

Suppose X satisfies

$$rX \quad \sigma X$$

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t$$

Divide the time interval $[0, T]$ into N parts: $\Delta t = T/N$, $t_n = n\Delta t$.

Define the m th simulated path by initializing $X_0^{(m)} = X_0$,

and given X_{t_n} , obtain $X_{t_{n+1}}$ by

$$X_{t_{n+1}}^{(m)} = X_{t_n}^{(m)} + a(X_{t_n}^{(m)}, t_n)\Delta t + b(X_{t_n}^{(m)}, t_n)Z_n^{(m)}\sqrt{\Delta t}$$

where $Z_n^{(m)}$ are IID standard normal. Evaluate the discounted payoff for the m th path. Take the average across all paths $m = 1, \dots, M$.

This extends directly to multidimensional state vectors X and multidimensional standard Brownian motion W .

Euler method convergence

With some assumptions on a and b , the Euler method has *weak* order of convergence 1 for general f , meaning

$$|\mathbb{E}f(X_T) - \mathbb{E}f(X_T^{(m)})| = O(\Delta t)$$

(Estimating $\mathbb{E}f(X_T^{(m)})$) produces additional error, not included here.) **skip**

Error analysis: Let $a_t = a(X_t)$ and $b_t = b(X_t)$. First subinterval:

$$\begin{aligned} X_{t_1} &= X_0 + \int_0^{t_1} a_t dt + \int_0^{t_1} b_t dW_t \\ &= X_0 + \int_0^{t_1} \left(a_0 + \int_0^t a \frac{\partial a}{\partial x} + \frac{1}{2} b^2 \frac{\partial^2 a}{\partial x^2} ds + \int_0^t b \frac{\partial a}{\partial x} dW_s \right) dt \\ &\quad + \int_0^{t_1} \left(b_0 + \int_0^t a \frac{\partial b}{\partial x} + \frac{1}{2} b^2 \frac{\partial^2 b}{\partial x^2} ds + \int_0^t b \frac{\partial b}{\partial x} dW_s \right) dW_t \end{aligned}$$

The Euler scheme keeps **three terms** to generate $X_{t_1}^{(m)}$

Crude analysis of discretization error of Euler scheme

- ▶ Intuition for weak error: $|\mathbb{E}(X_T - X_T^{(m)})|$ is $O(\Delta t)$, because there are N time steps, and $\mathbb{E}$ of error is $O(\Delta t)^2$ at each step (the only nonzero- $\mathbb{E}$ error term in previous equation is the $dsdt$ term).

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- ▶ Euler scheme has strong order of convergence $1/2$, meaning that

skip

$$\mathbb{E}|X_T - X_T^{(m)}| = O(\Delta t)^{1/2} \quad \text{as } \Delta t \rightarrow 0$$

Intuition for strong error: Biggest ignored term is $dW_s dW_t$.

Variance of error at one time step = $O(\text{Var}(\Delta W)^2) = O(\Delta t)^2$.

Variance of total error $X_T - X_T^{(m)}$ after N time steps = $O(\Delta t)$.

Standard deviation of $X_T - X_T^{(m)}$ after N time steps = $O(\Delta t)^{1/2}$.

This suggests that strong order of convergence is $1/2$.

Milstein scheme

Skp.

Milstein scheme: Don't ignore the term

$$\int_0^{t_1} \int_0^t b(X_s) \frac{\partial b}{\partial x}(X_s) dW_s dW_t.$$

Approximate it as

$$\begin{aligned} b(X_0) \frac{\partial b}{\partial x}(X_0) \int_0^{t_1} \int_0^t dW_s dW_t &= b(X_0) \frac{\partial b}{\partial x}(X_0) \int_0^{t_1} W_t dW_t \\ &= \frac{1}{2} b(X_0) \frac{\partial b}{\partial x}(X_0) (W_{t_1}^2 - t_1) \end{aligned}$$

So Milstein is

$$X_{t_{n+1}}^{(m)} = X_{t_n}^{(m)} + a(X_{t_n}^{(m)}) \Delta t + b(X_{t_n}^{(m)}) Z_n^{(m)} \sqrt{\Delta t} + \frac{1}{2} b \frac{\partial b}{\partial x}(X_{t_n}^{(m)}) ([Z_n^{(m)}]^2 - 1) \Delta t$$

Milstein has strong order of convergence 1, and weak order 1.

Weak convergence (important in option pricing): same order as Euler.

Covariance and correlation matrices

$$n \left[\begin{array}{c|c} & \text{Cov}(z_1, z_2) \\ \hline & \end{array} \right] m$$

Recall that the covariance matrix of a zero-mean vector Z is $\mathbb{E}(ZZ^\top)$.

Let M be a real symmetric matrix. The following are equivalent:

- ▶ M is a covariance matrix of some vector.
- ▶ M is *positive semi-definite*, which means that $x^\top Mx \geq 0$ for all real vectors x .
- ▶ The eigenvalues of M are all nonnegative.
- ▶ M has a *Cholesky decomposition* $LL^\top = M$ for some real lower-triangular matrix L with diagonal entries ≥ 0 .

$$dx_1 \quad dx_2$$

Covariance and correlation matrices

Moreover, the following are equivalent: $\text{Corr}(X, Y) = \text{Cov}\left(\frac{X}{\text{SD}(X)}, \frac{Y}{\text{SD}(Y)}\right)$

- ▶ M is a correlation matrix
- ▶ M is a covariance matrix and its diagonal elements are all 1.

Moreover, if M is a 3×3 matrix, the following are equivalent

- ▶ M is a correlation matrix.
- ▶ M is symmetric, its entries $\in [-1, 1]$, its diagonal entries $= 1$, and $\det M \geq 0$.

Interview question

Suppose $\text{Corr}(X, Y) = \text{Corr}(X, Z) = \text{Corr}(Y, Z) = \rho$.

What are the possible values of ρ ?

$$0 \leq \begin{vmatrix} 1 & \rho & \rho \\ \rho & 1 & \rho \\ \rho & \rho & 1 \end{vmatrix} = 1 + 2\rho^3 - 3\rho^2 = (\rho-1)(2\rho^2 - \rho - 1) \\ = (\rho-1)(\rho-1)(2\rho+1)$$

$$\begin{array}{rrrrr} 1 & 2 & -3 & 0 & 1 \\ & & 2 & -1 & -1 \\ \hline & 2 & -1 & -1 & 0 \end{array}$$

$$\Leftrightarrow 2\rho+1 \geq 0$$

$$\Leftrightarrow \rho \geq -\frac{1}{2}$$

Generating correlated Brownian motions

To get a D -dimensional vector $\bar{W}$ of BM with correlation matrix H , find a matrix $L \in \mathbb{R}^{D \times D}$ such that $LL^T = H$, and let $\bar{W} = LW$, where W is standard BM in $\mathbb{R}^D$.

So the process $dX_t = a(X_t, t)dt + b(X_t, t)d\bar{W}_t$ becomes

$$dX_t = a(X_t, t)dt + b(X_t, t)L dW_t$$

Simulation by Euler method:

$$X_{t_{n+1}}^{(m)} = X_{t_n}^{(m)} + a(X_{t_n}^{(m)}, t_n)\Delta t + b(X_{t_n}^{(m)}, t_n)L Z_n^{(m)}\sqrt{\Delta t}$$

where $Z_n^{(m)}$ are IID standard normal in $\mathbb{R}^D$. How to find L ?

- `numpy.random.Generator.multivariate_normal` generates LZ
- `numpy.linalg.cholesky` returns L given H

Generating correlated Brownian motions

Example: if we want $\bar{W} = \begin{pmatrix} \bar{W}^{[1]} \\ \bar{W}^{[2]} \end{pmatrix}$ with $\text{corr}(\Delta \bar{W}^{[1]}, \Delta \bar{W}^{[2]}) = \rho$, then

$$H = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

and Cholesky finds $LL^T = H$ where

$$L = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{pmatrix} \quad \begin{pmatrix} \bar{w}^1 \\ \bar{w}^2 \end{pmatrix} = L \begin{pmatrix} w^1 \\ w^2 \end{pmatrix}$$

$$\begin{aligned} d\bar{w}^{[1]} &= dw^{[1]} \\ d\bar{w}^{[2]} &= \rho dw^{[1]} + \sqrt{1-\rho^2} dw^{[2]} \end{aligned}$$

If Cholesky routine unavailable, can solve for L by traversing [the upper or lower triangular part of] H entry-by-entry. Each entry gives rise to an equation involving elements of L and only one unknown.

Generating correlated Brownian motions

Sometimes it is not necessary to simulate the entire path.

- ▶ Suppose $X_T - X_0$ is known to be multivariate normal with mean μT and covariance matrix HT .
 Dx1 DxD
- ▶ Suppose the option payoff depends only on X_T .

Then no need to divide $[0, T]$ into N steps. No need for Euler.

- ▶ Just generate

$$X_T^{(m)} := X_0 + \mu T + L\sqrt{T}Z^{(m)}$$

Dx1 Dx1 Dx1 DxD Dx1

where $LL^\top = H$ and the $Z^{(m)}$ are IID standard normal in $\mathbb{R}^D$.